

Forming Kinetics in HfO₂-Based RRAM Cells

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Abstract—In this paper, key aspects of the electroforming in resistive-RAM cells with HfO₂ are addressed. Specifically, transient measurements were performed on extremely short timescales to characterize systematically the forming time as a function of the voltage pulse amplitude, temperature, electrode materials, and device area. An analytical model was developed, accounting for the investigated experimental parameters. Impact of the forming (and SET) waveform on read disturb is also investigated.

Index Terms—Area scaling, electrodes, forming kinetics, forming pulse amplitude, HfO₂, MIM device, read disturb, resistive RAM (RRAM), temperature.

I. INTRODUCTION

AS WELL KNOWN, insulating films in integrated electronic devices can modify their behavior when subject to electric fields. Several electrical phenomena are observed depending on the external conditions: current leakage, reversible or irreversible forming of a conductive filament, and dielectric breakdown. In all those cases, a main macroscopic effect is a change of the film resistance. In the last years, the reversible resistance switching phenomenon has been attracting much attention, owing to the perspective of fast and dense nonvolatile-RAM technology based on the resistance change of transition metal oxide films in MIM devices [resistive RAM (RRAM)] (see, for example, [1]–[5]). The mechanism of forming in RRAM is not yet fully understood, although it appears evident that oxygen/vacancy motion [6]–[9], as well as material granularity [10], [11] and metal electrode interface [12], [13], plays a role.

In the last years, efforts have been devoted to the aim and a few models proposed in the literature [9], [14]–[17]. All models agree with the fact that the switching phenomenon is due to the formation and the rupture of a conductive filament driven by electric field or temperature or both. To get deeper insight into the kinetics of forming, it is necessary to investigate all the possible correlations with the parameters actually involved in the forming operation.

In this work, transient measurements were performed to investigate at the forming timescale the forming kinetics in

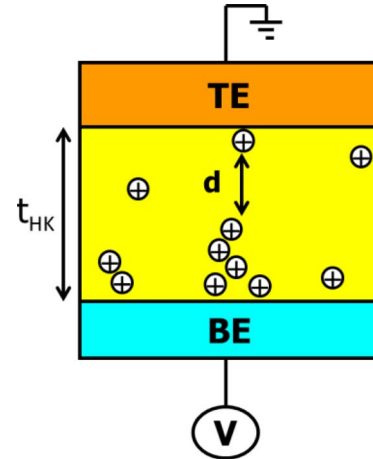


Fig. 1. Sketch of an oxygen vacancy distribution during the forming pulse.

cells with HfO₂ films. Specifically, the time for forming was systematically characterized as a function of the voltage pulse amplitude, temperature, electrodes, and device area.

A model of the forming time was proposed, which successfully fits all the data, and the impact of different forming and SET waveforms on the robustness of the low-resistance state (LRS) was also studied after constant voltage stress.

Samples of the types Pt/HfO₂/Pt (S1), TiN/HfO₂/Pt [S2, with TiN as bottom electrode (BE)], and TiN/HfO₂/TiN (S3) were investigated. HfO₂ was ALD deposited on sputtered BE. The top electrode (TE) was sputtered too, and the stack was patterned via hard-mask etching. The device area is 0.125 μm² (in all the experiments apart from that in the section of area scaling). X-ray diffraction revealed prevalence of the orthorhombic phase when the HfO₂ film was grown on TiN and the monoclinic phase when the HfO₂ film was grown on Pt. The HfO₂ thickness was 10 nm, and the Pt and TiN thickness was 25 nm. As expected, a film with an increased density of oxygen vacancies was detected by TEM with EDX at the bottom interface in the S2 and S3 samples [18], due to the reactivity of TiN with HfO₂. Other details on sample preparation and material analysis are reported in [18].

II. MODEL OF FORMING

As already mentioned, electroforming corresponds to the closure of a filament of traps connecting one interface to the other. In Fig. 1, traps are oxygen vacancies (circles with a cross inside) assumed to be positively charged, following [19]–[21].

The maximum time to complete a path (τ_{MAX}) is given by the time spent by an oxygen vacancy located at one interface to reach the other interface, through the whole oxide

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thickness (t_{HK}). Referring to the sketch in Fig. 1, τ_{MAX} can be written as

$$\tau_{\text{MAX}} = \alpha \frac{t_{\text{HK}}}{v_i} = \alpha \frac{t_{\text{HK}}}{\mu_i F_{\text{MIN}}} = \alpha \frac{(t_{\text{HK}})^2}{\mu_i V_P} \quad (1)$$

where v_i is the ion velocity, μ_i is the ion mobility, F_{MIN} is the minimum electric field in the oxide when the voltage V_P is applied, and α is a parameter accounting for nonstraight paths ($\alpha \geq 1$). Relying on the Einstein relation, the following holds:

$$\mu_i = \frac{q}{kT} D_0 \cdot e^{-\frac{E_A}{kT}} \quad (2)$$

where T is the temperature, k is the Boltzmann constant, D_0 is the diffusivity, and E_A is the hopping activation energy. However, in general, forming takes less time than τ_{MAX} since traps need to cover shorter distances. In fact, traps can be present in the oxide bulk (not only at the interfaces) and progressively pile up one above the other, forming a portion of the filament. As shown in Fig. 1, where d is the remaining distance to be covered ($d < t_{\text{HK}}$) to close the filament, it is worth noticing that the distance d varies with time during the filament closure and definitely with the local field itself. As d decreases, the local field F grows accordingly

$$F = \frac{V_P}{d} > F_{\text{MIN}}. \quad (3)$$

As a result, the actual forming time (τ_F) is lower than τ_{MAX} and depends on the forming conditions. We may express this dependence by introducing a dimensionless parameter f , whose numerical value is lower than unity in any condition and whose dependence on the forming conditions is unknown

$$\begin{aligned} \tau_F &= f \cdot \tau_{\text{MAX}} \\ \tau_F &= \alpha \cdot \frac{(t_{\text{HK}})^2}{qD_0} \cdot \frac{kT}{V_P} \cdot f \cdot e^{\frac{E_A}{kT}}. \end{aligned} \quad (4)$$

The equation above includes the possible impact of the forming conditions on the ion diffusivity. In the next section, the analytical expression of f will be extracted from the comparison of (4) with experiments.

III. EXPERIMENTS AND DISCUSSION

Preliminary characterization of forming, SET, and RESET and a short cycling were performed in quasi-static (QS) condition using an Agilent B1500 semiconductor parameter analyzer (sweep speed of 1 V/s). Forming and switching are shown in Fig. 2 for samples S1, S2, and S3, respectively. The average values of the forming voltage (V_F), RESET voltage (V_{RESET}), and SET voltage (V_{SET}) were 4, 0.7, and -0.8 V, respectively, for the S2 and S3 samples (bipolar behavior).

In Fig. 2 (right side), the switching behavior for a limited number of cycles is shown. The resistance values were obtained from I - V measurements. In order to reduce fluctuations, values were averaged in the range between 10 and 100 mV.

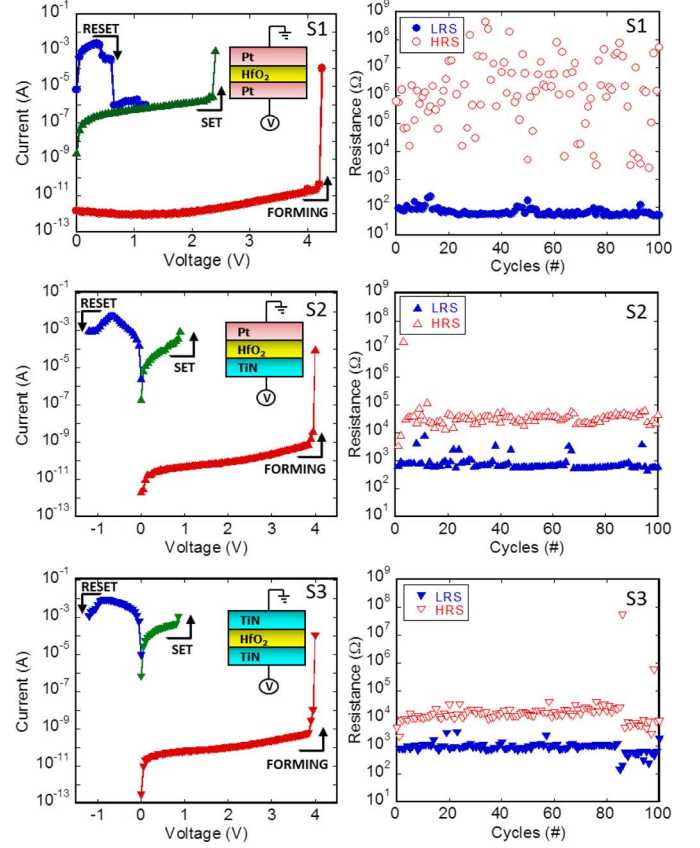


Fig. 2. (Left) Forming, SET, and RESET curves of the S1, S2, and S3 samples. (Right) Switching behavior of the high- and low-resistance states (average value measured between 10 and 100 mV) of the S1, S2, and S3 samples.

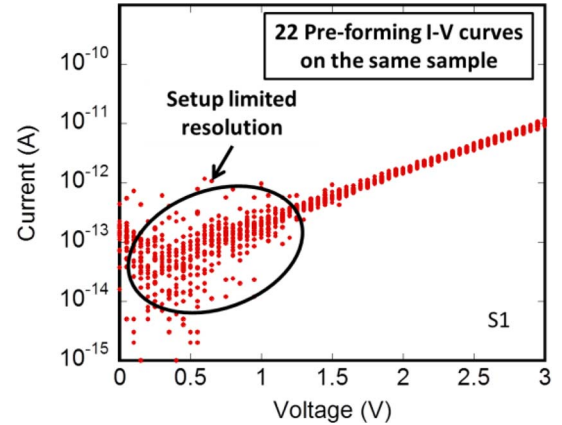


Fig. 3. Twenty-two QS I - V curves performed consecutively on the same S1 sample stopped just before forming.

A. Tradeoff Relationship Between Fast Forming and Low Programming Voltage and Role of the Electrode Materials

The dependence of forming on voltage was investigated in QS and transient conditions. In QS condition, several ramps were applied consecutively on the same sample, always stopping just before the minimum value of the forming voltage distribution, and the I - V curves were acquired.

As a result, the I - V curves were superimposed (an example is shown in Fig. 3). This indicates that there is not a cumulative effect in the filament forming but, rather, a threshold behavior with voltage.

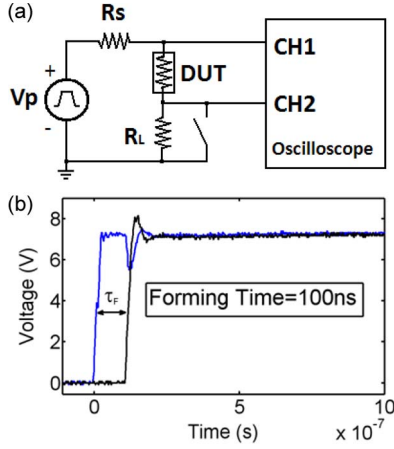


Fig. 4. (a) Sketch of the circuit used for measuring forming time and resistance. (b) Waveforms detected on CH1 and CH2 of the oscilloscope.

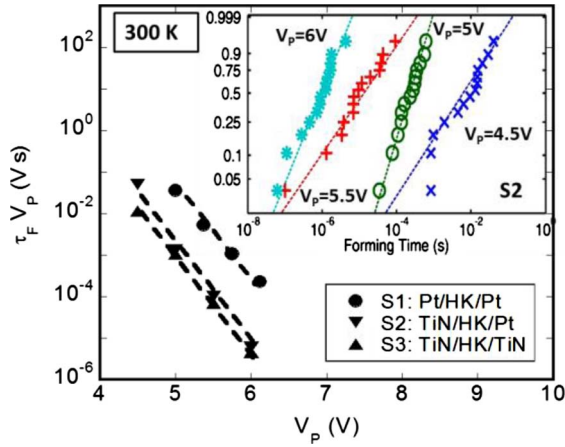


Fig. 5. Data of forming time (average value) varying the voltage pulse amplitude for the S1, S2, and S3 samples. The markers are experimental data; the lines are interpolations. (Inset) Weibull distributions of one set of samples (S2).

To get deeper insight, transient measurements were performed at extremely short times. Voltage pulses with 10-ns rise time (T_{RISE}) and amplitude (V_P) greater than the average V_F value found in QS condition were used. The circuit used in this experiment is shown in Fig. 4(a). Pulses were displayed on channel 1 (CH1) of an oscilloscope. The voltage drop on a resistor in series to the device under test (DUT) was acquired on channel 2 (CH2) of the oscilloscope. $R_L = 10 \text{ k}\Omega$ fixed the current compliance to $V_P/10 \text{ k}\Omega$.

The typical waveforms acquired on CH1 and CH2 are shown in Fig. 4(b). The signal displayed on CH2 allowed measuring τ_F .

Data of τ_F for the S1, S2, and S3 samples were collected at room temperature. The average value τ_F was then used to plot $V_P \cdot \tau_F$ against V_P in Fig. 5. The following analytical behavior is found for all the samples (the dashed lines in Fig. 5):

$$\tau_F \propto \frac{e^{-\gamma V_P}}{V_P} \quad (5)$$

with $\gamma = 6 \text{ V}^{-1}$, regardless of the electrodes. Comparing (5) with (4), it comes out that $f(V_P) = \exp(-\gamma V_P)$. As a result,

we say that voltage scaling is limited by the tradeoff with the forming time. As an example, in sample S1, the characteristic time of forming is $200 \mu\text{s}$ at 6.12 V and grows up to two orders of magnitude with a voltage decrease of only 18% (30 ms at 5 V). The parameter relating the forming time with voltage in the exponential term [γ in (5)] is the most important for the fast programming with low voltage. However, the experimental datum of γ (6 V^{-1}) is rather small and does not depend on the electrode material.

We remark here that the QS measurements featured 100-ms hold times during the sweep, and for this reason, in Fig. 2, forming occurred at lower voltage values with respect to transient conditions and shown in Fig. 5. It is worth noticing that progressive electroforming process could not be appreciated in the quasi-stationary measurements of Fig. 3 where the sweep times were very much shorter than the ones extrapolated from Fig. 5 in that voltage range (thousands of seconds).

The forming times for the S2 samples (featuring TiN as BE and Pt as TE) and S3 samples (featuring TiN as BE and also as TE) are approximately the same (Fig. 5). At the same time, the S2 and S3 curves lie at times shorter with respect to the S1 curve (featuring Pt as BE and TE). This indicates that the chemical reactivity of the growth electrode (rather than the TE) plays a main role in the forming kinetics. The measured shorter τ_F in the case of TiN as BE can be explained with the formation of a TiON layer with consequent enrichment of oxygen vacancies at the bottom interface, which accelerates creation of percolative paths [12].

B. Role of Temperature

The chuck temperature (T) was increased up to 473 K , and V_P was varied. The data of τ_F/T versus $1000/T$ (and V_P) are graphed in Fig. 6 for samples S1, S2, and S3. All the collected data were successfully interpolated with the model

$$\tau_F = \alpha \frac{(t_{\text{HK}})^2}{qD_0} \frac{kT}{V_P} e^{-\gamma V_P} e^{\frac{E_A}{kT}} \quad (6)$$

using $\alpha = 1$, $\gamma = 6 \text{ V}^{-1}$, and $D_0 = 3.3 \cdot 10^{-15} \text{ cm}^2/\text{s}$. Interpolations are shown in the figure with dashed lines. The only parameter which varied with the growth electrode was the hopping activation energy. The average values of E_A extracted from the fits are 0.72 eV for the S1 samples and 0.56 eV for the S2 and S3 samples.

The hopping energy for samples with a bottom interface layer rich of oxygen vacancies is lower than that for samples with Pt electrode.

C. Area Scaling

The forming time was monitored, with respect to device area (A) in the range of $0.125\text{--}2.544 \mu\text{m}^2$. This investigation was performed only on nonpolar S1 samples. The results (each point is an average of 20 samples) are shown in Fig. 7 for different values of the pulse amplitude. As a result, it was found that, above a critical area (A_C), τ_F tends to saturate. The asymptotic value of τ_F is higher at lower values of V_P ,

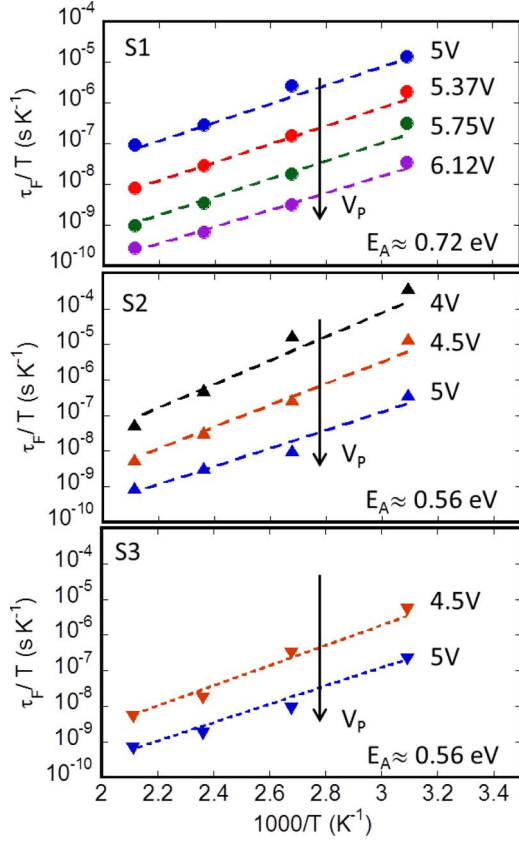


Fig. 6. Data of forming time varying the temperature for the S1, S2, and S3 samples. The markers are experimental data; the lines are interpolations.

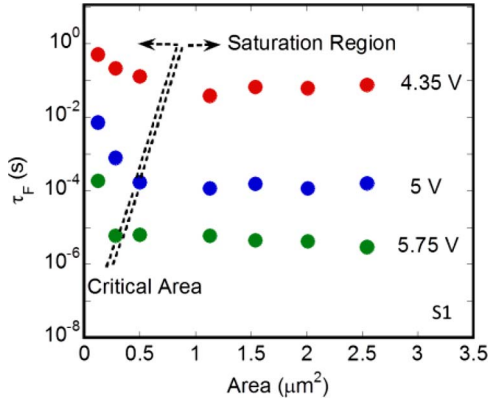


Fig. 7. Forming time versus area for three pulse amplitudes (S1 samples).

which is in agreement with previous results. Below A_C , we find the expected behavior: The smaller the area, the lower the number of traps, and the higher the time required for the filament forming. A_C is inversely related to V_P . This fact is a concern for RRAM device scaling, since the contemporary scaling of area and voltage implies a strong increase of forming time.

The proposed model accounts for the area scaling if a dimensionless factor (η_A , varying with area) is included in the function f

$$\tau_F = \alpha \cdot \eta_A \cdot \frac{(t_{HK})^2}{qD_0} \frac{kT}{V_P} e^{-\gamma V_P} e^{\frac{E_A}{kT}}. \quad (7)$$

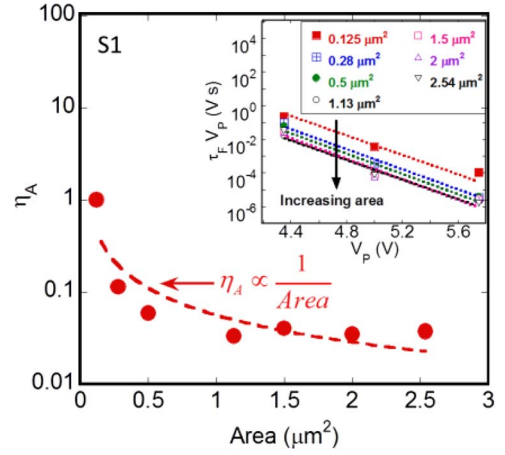


Fig. 8. Values of the normalized pre-exponential factor η_A which allow fitting area scaling with (7) (the fits are shown in the inset; the areas are in square micrometers).

The data in Fig. 7 were interpolated with (7), setting $E_A = 0.72$ eV (S1 samples), $D_0 = 3.3 \cdot 10^{-15}$ cm²/s, $\gamma = 6$ V⁻¹, $\alpha = 1$, and $T = 300$ K and varying only the value of η_A . The extracted values of η_A are shown in Fig. 8 as a function of the area, and the relative fits are shown in the inset as a function of V_P .

The value of η_A is one for the 0.125-μm² cell since it was the reference area in the experiments in the previous sections, from which the value of D_0 was extracted.

The behavior of η_A with the area obviously follows the behavior of τ_F with the area. An inverse relation of the type

$$\eta_A \propto \frac{1}{\text{Area}} \quad (8)$$

is displayed with the dashed line in the figure. That interpolation could be used to some extent as a predictive model. To explain the behavior of τ_F at large areas, we can consider that, after increasing the area, the probability to form a filament tends to unity. Probably, more filaments add in parallel, but this has no impact on the time for forming. In the investigated case (Pt electrodes), the values of A_C are greater than what is foreseen by the ITRS for the next node of nonvolatile memory, which implies that the region of interest for practical applications is in the left side of Fig. 7, where the forming time increases with area scaling. The dependence of forming voltage on the cell size in a 5-nm-thick Ti/HfO₂/TiN MIM device was reported in [22]. As a result, in that work, a drastic increase of breakdown voltage was found below a characteristic size. This is in complete agreement with our findings regarding forming times.

D. Impact of Forming/SET Waveform on Read Disturb

In this experiment, forming (and SET) was performed on different S1 samples with different waveforms, and the robustness of the LRS to read disturb was investigated in three conditions.

The three used waveforms are shown in Fig. 9. The current compliance was the same in the three cases (600 μA). Waveform W1 was a long ($T_P = 200$ μs) pulse with $V_P = 6$ V and very short T_{RISE} (10 ns). Waveform W2 was a long-lasting

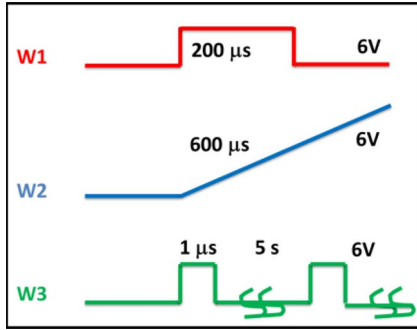


Fig. 9. Sketch of the three waveforms used for forming and SET in the read disturb experiment on S1 samples.

ramp ($600 \mu\text{s}$) with maximum value of 6 V. In the case of W1, forming took place during the pulse; in the case of W2, forming took place during the ramp. The third waveform (W3) was made of a train of short pulses (N_P) featuring $T_{\text{RISE}} = 10 \text{ ns}$, $T_P = 1 \mu\text{s}$, $V_P = 6 \text{ V}$, and very long time between two pulses ($T_{\text{OFF}} = 5 \text{ s}$). Due to the distribution of τ_F values, the resistance was measured during T_{OFF} to verify if the conductive filament had closed with the previous pulse. In that case, the pulse train was interrupted. Therefore, N_P was the minimum number of pulses guaranteeing to obtain forming with the distribution shown in the inset of Fig. 5. After forming, a RESET and a subsequent SET were performed. RESET was performed with 2-V 1-ms pulses for all the samples, and SET had the same features as forming. W1 was the only waveform guaranteeing the following: 1) The conductive filament closed in stationary condition, and 2) the traps in the conductive path had time enough to settle and rearrange. A set of S1 samples (30) was investigated.

The measured distributions of resistance values after SET operation are shown in Fig. 10. As we can see, the spread is rather small in all the cases (50–80 Ω), and for W3, the distribution is the most narrow with the highest median resistance value. After, the samples were stressed applying a constant voltage (CVS at 0.3 V) much lower than the RESET voltage, and the values of the LRS were monitored during the stress. The CVS was interrupted when LRS had increased by a factor of ten. We define that time as failure time (τ_{FAIL}). In the inset of Fig. 11, an example of the measured LRS during the stress is shown with the $10\times$ increase outlined.

The measured distributions of τ_{FAIL} are shown in Fig. 11. As a result, extremely different distributions of τ_{FAIL} were found, depending on the waveform. In particular, W3 samples exhibit the most narrow distribution but, at the same time, the shortest life against read disturbance. On the contrary, W1 seems to guarantee higher robustness with respect to read disturbance but introduces extremely broad tails.

Furthermore, in the case of W1 and W2, the voltage applied during forming continues to stress the material, and the compliance current continues to flow for long times after the filament formation. Therefore, we can expect that, with W1 and W2, the formed filament undergoes evolutions, which can vary from one sample to another depending on the microscopic features and the local temperature, possibly increasing their medium size and/or creating more than one conductive path. This intuitively

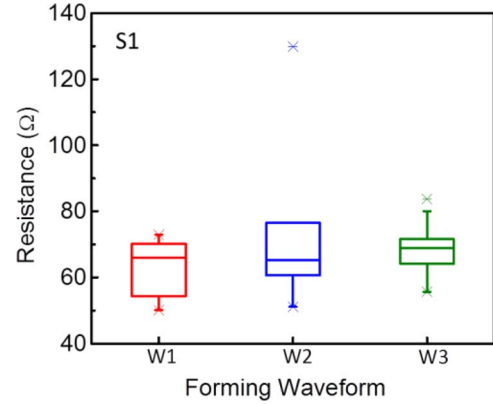


Fig. 10. Distributions of resistance values after SET operations performed with W1, W2, and W3, before the read disturb test.

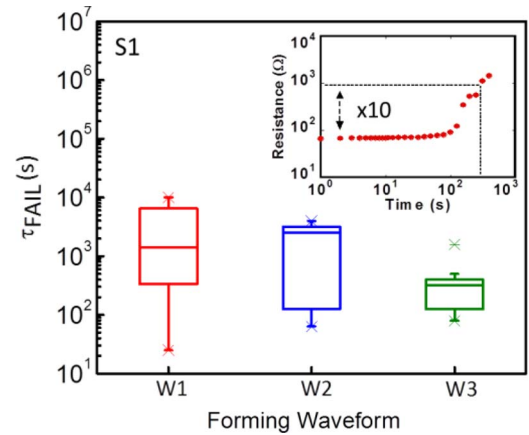


Fig. 11. Failure time values of S1 cells subjected to forming (and SET) waveforms W1, W2, and W3 after a constant read disturb test at 0.3 V. Data collected on 30 samples. (Inset) Example of the LRS value with the $10\times$ increase, defined as the failure time criterion.

gives a wider distribution of resistance in the case of longer stress, than with W3. Furthermore, looking at Figs. 10 and 11, we can state that W3 gives the highest median value of resistance before stress and the shortest lifetime. At the same time, W3 implies the less dispersed distributions of failure time and resistances. A tradeoff between failure time median and distribution is desired.

IV. CONCLUSION

Systematic characterization of forming in RRAM MIM cells with HfO_2 films has been addressed. Specifically, forming kinetics has been investigated at extremely short times in a wide range of different experimental conditions.

An analytical model has been proposed, which accounts for the influence of the voltage pulse amplitude, temperature, area, and electrodes on the forming time. The only parameter to be changed to fit all the collected data was the ion hopping activation energy, which varied with the electrode. In particular, a lower value of the activation energy was the reason of the shorter forming times in samples with TiN electrode. In fact, TiN reacting with the high- k film gives rise to the formation of an interface layer rich of oxygen vacancies, which lowers the ion hopping activation energy and decreases the forming times with respect to nonreacting Pt electrodes.

The forming time increases with voltage scaling and, below a critical area, also with area scaling. In turn, the value of the critical area depends on the forming voltage. This would be a concern, in the sense that scaling of area increases the forming time. Finally, the impact of different forming and SET waveforms on the robustness to voltage stress (of low entity) has been investigated. We conclude that waveforms ensuring the following guarantee higher robustness to read disturb: 1) The conductive filament forms in stationary condition, and 2) the traps in the path have time to settle and rearrange. However, this is paid in terms of broader distributions of the failure time.

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